



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

AUTOCAD AND COVADIS SOFTWARE IN SURVEYING PROJECT

LSVAC402

Apply AutoCAD and Covadis Software in Surveying Project

Competence

RQF Level: 4

Learning Hours



100

Credits: 10

Sector: Construction and Building Services

Trade: Land Surveying

Module Type: Specific

Curriculum: CBSLSV4001 – TVET Certificate IV in Land Surveying

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Issue Date: May 2023

Purpose statement	This module describes the knowledge, skills, and attitudes required to apply AutoCAD and Covadis Software. It is designed for students pursuing RQF level 4 in Land Surveying. At the end of this module, the students will be able to install the AutoCAD and Covadis software, configure AutoCAD and Covadis user interface, produce AutoCAD and Covadis drawings and plot AutoCAD and Covadis drawings.					
Learning assumed to be in place	Technical drawings interpretation, computer literacy, French, English, AutoCAD basic skills, earthwork computation, surveying computation and adjustment.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group work and presentation	30%				
	• Individual work	40%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Install AutoCAD and Covadis software	1.1. Specification and operating system of computers are appropriately identified according to their types.
	1.2. AutoCAD and Covadis software are properly selected according to the version and computer's specifications.
	1.3. AutoCAD and Covadis software are successfully installed according to their version.
	1.4. AutoCAD and Covadis software are successfully activated based on the version.
2. Configure AutoCAD and Covadis user interface	2.1. AutoCAD and Covadis workspace components are effectively identified according to the software version.
	2.2. AutoCAD and Covadis toolbars are adequately activated according to their uses.

	2.3. Drawing properties are accurately set according to the project standards.
3. Produce AutoCAD and Covadis drawings	3.1. 3D models are accurately created according to the project requirements.
	3.2. Topographic maps are accurately created according to the project requirements.
	3.3. Cadastral plans are accurately created according to the project requirements.
	3.4. Profiles are accurately produced according to their types.
	3.5. Earthwork is accurately computed according to the software procedures.
4. Plot AutoCAD and Covadis Drawings	4.1. Annotation, labels and hatches are appropriately added inline to the drawing requirements.
	4.2. Drawings are appropriately organized into layers according to their convention presentation.
	4.3. Plotting layout is properly designed according to the drawing content and agreed formats.
	4.4. Earthwork is accurately computed according to the calculation methods selected.

Intended Knowledge, Skills, and Attitudes

Knowledge	Skills	Attitude
✓ Identify computer specifications and operating systems ✓ Describe the 3D models ✓ Identify Autocad and Covadis installation requirements ✓ Identify Covadis tools ✓ Identify layout ✓ Perform earthwork computation using Autocad and Covadis ✓ Identify technical drawings ✓ Interpret technical drawings	✓ Install Autocad and Covadis software ✓ Perform Autocad and Covadis basic settings ✓ Organise drawings into layers ✓ Create design layout ✓ Produce topographic maps ✓ Produce mass haul diagrams ✓ Create profiles ✓ Perform Printing	✓ Use analytical skills to draw mass haul diagrams ✓ Use creativity and innovation throughout the design works ✓ Pay attention to projects details ✓ Demonstrate punctuality during area and volume field data collection ✓ Demonstrate resourcefulness in mass haul diagrams calculations, drawing and interpretation

Course content

Learning outcomes	At the end of this module the learner will be able to: 1. Install AutoCAD and Covadis software 2. Configure AutoCAD and Covadis user interface 3. Produce AutoCAD and Covadis drawings 4. Plot AutoCAD and Covadis Drawings
Learning outcome 1: Install AutoCAD and Covadis software	Learning hours: 10
Indicative content	
• Identification of specifications and operating systems of computer ✓ Types of operating system of computer  32 bytes  64 bytes ✓ Specifications of computer  Processor  Memory (RAM)  Windows version • Selection of AutoCAD and Covadis software ✓ Software version ✓ Software requirements • Installation of AutoCAD and Covadis software ✓ Steps of software installation • Activation of AutoCAD and Covadis software application ✓ Identification of activation key ✓ Application of activation key	
Resources required for the learning outcome	
Equipment	Computer, projector, white-board.
Materials	Marker pen, duster, flip chart, chalk, paper.

Tools	AutoCAD and Covadis Software, storage devices.
Facilitation techniques	Trainer guided, Individual and group work, Theoretical and Practical exercises.
Formative assessment methods /(CAT)	Written assessment, Oral assessment, Performance assessment, Product based assessment.

Learning outcome 2: Configure AutoCAD and Covadis user interface	Learning hours: 20
Indicative content	
• Identification of AutoCAD and Covadis workspace components ✓ AutoCAD classic ✓ 3D Modeling • Activation of AutoCAD and Covadis toolbars ✓ AutoCAD toolbars ✓ Covadis toolbars • Setting of Drawing properties ✓ Layer tools ✓ Color ✓ Line type ✓ Line weight ✓ Scale list ✓ Text style ✓ Dimension style ✓ Table style ✓ Point style ✓ Multiline style ✓ Drawing Units	
Resources required for the learning outcome	

Equipment	Computer, projector, white-board.
Materials	Marker pen, duster and, flip chart, chalk, papers.
Tools	AutoCAD and Covadis Software, storage device.
Facilitation techniques	Trainer guided, Individual and Group work, Theoretical and Practical exercises
Formative assessment methods /(CAT)	Written assessment, Oral assessment, Performance assessment, Product based assessment.

Learning outcome 3: Produce AutoCAD and Covadis drawings	Learning hours: 55
Indicative content	
• Creation of 3D modelling ✓ 3D modelling annotation ✓ 3D modelling architectural ✓ 3D modelling Mechanical ✓ 3D modelling electrical ✓ 3D modelling civil/structure ✓ 3D modelling hatches • Creation of Topographic maps ✓ Identification of data formats ✓ Steps of data display ✓ Create TIN (Triangular Irregular Networks) ✓ Produce contour lines  Methods of creating contour lines (Semi-automatics and Automatics)  Steps of contour lines creation ✓ Configuration of contour lines ✓ Creation of features ✓ Produce topographic maps • Creation of Cadastral plans	

✓ Identification of data formats ✓ Creation of layers ✓ Data display ✓ Drawing of features ✓ Computation of distances and bearings ✓ Produce cadastral plan layout • Producing profiles ✓ Creation of an alignment/ centreline ✓ Longitudinal profile ✓ Cross sectional profile • Computation of Earthwork ✓ Areas ✓ Volumes	
Resources required for the learning outcome	
Equipment	Computer, projector, white-board, total station, GNSS Receiver, PPE.
Materials	Marker pen, duster, flip chart, chalks, papers.
Tools	AutoCAD and Covadis Software, existing data, storage devices.
Facilitation techniques	Trainer guided, Individual and group work, Theoretical and Practical exercises.
Formative assessment methods /(CAT)	Written assessment, Oral assessment, Performance assessment. Product based Assessment.

Learning outcome 4: Plot AutoCAD and Covadis Drawings	Learning hours: 15
Indicative content	
• Creation of annotation, labels and hatches ✓ Insert text ✓ Dimensions ✓ Hatches and gradient	

- **Organisation of drawings into layers**
 - ✓ Working with layers
 - ✓ Layer properties
 - ✓ Presentation of drawings into layers
- **Designing plotting layout**
 - ✓ Creating and updating drawing layout
 - ✓ Designing of title block
 - ✓ Arrangement of drawings
- **Plotting and exporting of drawings**
 - ✓ Procedures of plotting
 - ✓ Procedures for exporting

Resources required for the learning outcome

Equipment	Computer, projector, white-board.
Materials	Marker pen, duster, flip chart, chalks, papers.
Tools	AutoCAD and Covadis Software, geospatial data, storage device.
Facilitation techniques	Trainer guided, Individual and group work, Theoretical and Practical exercises.
Formative assessment methods /(CAT)	Written assessment, Oral assessment, Performance assessment, Product based assessment.

Integrated/Summative assessment

Integrated situation

Kayonza District is committed to developing sustainable infrastructure for all sectors starting from establishing the road network. The district has already called upon professionals in land surveying to collect data on field and produce maps, drawings and plans required for project cost estimation. GDEA Company Ltd, experienced in road geometrical design is awarded the tender. As a CAD Draftsman assistant hired by the company, you are requested to perform the tasks specified as follow:

- To produce drawings relevant for the construction of the 9km unpaved road Abanyonzi- Ruramira in Ruramira Sector.
- To process data using AutoCAD and Covadis software
- To produce layout plan
- To produce longitudinal profile with maximum gradient $\pm 7\%$
- To produce typical cross-section profile each at 20m interval with cross fall slope with $\pm 3\%$

The scales of 1/1000 for horizontal and 1/250 for vertical should be used for longitudinal profile.

The data to be used is available in the office of GDEA Company Ltd and includes:

- Points ID for centreline points indicated, Eastings, Northings, Elevations and description of survey points crossing each chainage.
- The location of the existing road features is also indicated.
- These tasks must be done in 12 hours using AutoCAD and Covadis software.
- The submission folder should include the whole project (original ground data, layout plan, longitudinal profile, cross section profile, typical cross sections, table of earthworks and setting out data) in editable dwg and pdf formats.

Resources

Tools	AutoCAD and Covadis Software, existing data, storage devices.
Equipment	Computer, projector, white-board.

Materials/ Consumables		Marker pen, duster, flip chart, chalk, paper.			
Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
1. Install AutoCAD and Covadis software (20%)	1.1. Specification and operating system of computers are appropriately identified	1. AutoCAD and Covadis 2007 are properly selected			2
		2. Computer Hardware is properly selected			2
		3. Specifications of computer are well identified			2
		4. Operating system of computer is well identified			2
	1.2. AutoCAD and Covadis software	1. Covadis software is well selected			2
		2. AutoCAD software is well selected			2
	1.3. AutoCAD and Covadis software are successfully installed	1. The steps for software installation are well respected			4
	1.4. AutoCAD and Covadis software are successfully activated	1. AutoCAD software is activated			2
		2. Covadis software is activated			2
	2. Configure AutoCAD and	2.1. AutoCAD and Covadis workspace	1. AutoCAD classic is activated		

Covadis user interface (20%)	components are effectively identified				
	2.2. AutoCAD and Covadis toolbars are adequately activated	1. AutoCAD toolbars are activated			2
		2. Covadis toolbars are activated			2
	2.3. Drawing properties are accurately set	1. Line /polyline are used			2
		2. Dimension feature is used			4
		3. Text feature is used			4
		4. The metric unit is used			4
3. Produce AutoCAD and Covadis drawings 40%	3.1. D models are accurately created	1. Hatch features are used			2
		2. Gradient features are used			2
	3.2. Topographic maps are accurately created	1. Data formats are respected			2
		2. Data are properly processed			3
		3. Data are correctly checked			3
	3.3. Cadastral plans are accurately created	1. Layers are accurately created			3
		2. Coordinates are accurately displayed			2
		3. Drawing of features is well performed			2
		4. Distances and bearings are computed correctly			2
		5. Cadastral plan layout			2

		well designed			
	3.4. Profiles are accurately produced	1. Cross fall slope with $\pm 3\%$ is accurately respected			5
		2. Profiles of 9km are well produced			2
		3. Cross sections of 7m are well produced			3
		4. Chainage at 20m is well respected			3
	3.5. Earthwork is accurately computed	1. Procedures of earthwork are respected			4
4. Plot AutoCAD and Covadis Drawings (20%)	4.1. Annotation, labels and hatches are appropriately added	1. Annotation is well indicated			2
		2. Labels are well presented			2
		3. Hatches are well indicated			2
	4.2. Drawings are appropriately organized into layers	1. The steps of plot layout design are well respected			2
		2. The existing road features are well indicated.			2
	4.3. Plotting layout is properly designed	1. Horizontal scale of 1:1000 is well respected			2
		2. Vertical scale of 1:250 is well respected			1
		3. Pdf format is produced properly			1

		4. Editable or dwg format is produced correctly			2
		5. Maximum gradient of ±7 is well respected			2
		6. Time is respected			2
Total marks		100%			
Percentage Weightage		100%			

References

1. Alain j.kalameja, d. c. (2009). *AutoCAD 2010 Tutor for Engineering Graphics*. Delhi.
 2. KR.Gopalakrishna. (1991). *Engineering Drawing*.
 3. s.a, G. (1993). *Logiciel de topographie et projets VRD; Le centre des systemes agrees AutoCAD en topographie, Cartographie eS.I.G.*
 4. Venugopal, K. (2004). *Engineering drawing and graphics +AutoCAD*. New Delhi: New Age.
 5. Wilson, R. (1990). *Engineering Drawing and Design*. New York: Edward Arnold.
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